HSAN ELLOUZE

SOFTWARE ENGINEERING STUDENT | FULLSTACK & DEVOPS ENGINEER | AI SYSTEMS

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SUMMARY

Software Engineering student specializing in Fullstack development, DevOps, and AI systems. Experienced in building scalable applications, designing CI/CD pipelines, and developing intelligent systems using modern cloud-native technologies. Passionate about Kubernetes, observability, and distributed systems.

EXPERIENCE

Software Engineer Intern
07/2025 - 09/2025
Mazdev

- Engineered production chatbot platform with YAML configuration enabling non-technical users to create 20+ bot variants.
- Developed RAG engine using Pinecone achieving 94% semantic accuracy across 50K+ embedded documents.
- Implemented automated testing framework for 150+ scenarios reducing QA cycle by 50%.
- Optimized embeddings reducing latency by 45% for user-facing interactions.

Fullstack Developer
03/2025 - 06/2025
Mazdev

- Architected AI-powered legal contract review platform for 40+ enterprise clients reducing analysis time by 60%.
- Built full-stack system (React/Next.js + FastAPI) processing 500+ monthly contract reviews with 100K+ MongoDB documents.
- Implemented Docker CI/CD pipeline reducing infrastructure setup time by 75%.
- Designed optimized data models supporting large-scale document storage.

Web Developer Intern
08/2024 - 09/2024
Steros

- Designed responsive website generating 30% increase in service inquiries.
- Integrated real-time notifications improving engagement by 25%.
- Built MySQL schema supporting multi-tenant data separation with zero-downtime migration.

ACADEMIC PROJECTS

Kubemind
03/2026 -
AI-powered Kubernetes Observability Platform

- Designed and implemented an AI-driven observability platform for Kubernetes, integrating logs, metrics, and events
- Built a real-time data ingestion pipeline using Prometheus, Loki, and Kafka
- Developed modular AI agents for anomaly detection across logs, metrics, and cluster events
- Implemented a correlation engine to detect incidents by combining multi-source signals and deployed the system on Kubernetes

Technologies: Kubernetes, Docker, Prometheus, Loki, Kafka, Python, FastAPI

PromptCraft
05/2025 - 06/2025
AI-Powered NFT Generation & Auction Platform

- Built a full-stack AI-powered NFT generation and real-time auction platform supporting 100+ concurrent users.
- Implemented WebSocket and SSE-based bidding system reducing update latency by ~60%.
- Developed scalable NestJS REST and GraphQL APIs handling 1K+ daily requests.

Technologies: Vue, Velzon, NestJS, WebSocket, GraphQL, REST API, SSE

WEBSITES & SOCIAL

Portfolio: hsanellouze.me
LinkedIn: linkedin.com/in/hsanellouze
GitHub: github.com/Hsan003

EDUCATION

INSAT - National Institute of Applied Sciences and Technology
2022
B.S. Engineering in Software Engineering
EQF Level 7. Specializations: Full-Stack Development, AI Systems, Cloud Architecture.

SKILLS

Languages
Python | JavaScript | SQL

Frontend
React | Next.js | Vue

Backend
FastAPI | NestJS | ASP.NET Core

DevOps & Cloud
Docker | Kubernetes | GitHub Actions | CI/CD

Observability & Data
Prometheus | Loki | Kafka | MongoDB

Tools
Git | Linux

ACTIVITIES

EVENT MISSION LEADER
Junior Enterprise INSAT
Aug 2024 - Jun 2025

- Orchestrated INSAT Enterprise Forum 6.0, coordinating 30+ corporate partners for 1500+ participants.
- Led 40+ organizers through strategic planning and resource allocation.

OPERATIONS MANAGER
WinterCup 7.0 | ACM INSAT
Feb - Mar 2025

- Managed operations for national programming competition attracting 150+ participants.
- Developed procedures and improved volunteer coordination efficiency.
- Maintained competition integrity across technical checkpoints.